

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

Grade 6 Mathematics

Term 1 — Lesson Plans (Lesson Exemplars)

SY 2026–2027 · MATATAG

Number & Algebra and an introduction to Geometry
4 A's format · aligned to the official Budget of Work

Lesson Plans

Six lesson exemplars covering Term 1 competencies

Each lesson runs 1–2 sessions (50 min each). Procedure follows Activity → Analysis → Abstraction → Application.

Lesson 1 · Operations on Decimals NA 6NA-Ia-1 / 6NA-Ib-2

Subject Matter	The four fundamental operations on decimals; multi-step word problems
Time	2 sessions (100 min)
Materials	place-value chart, play money (peso bills/coins), grocery receipts, manila paper
References	MATATAG Math CG Grade 6; Term 1 Budget of Work

LEARNING OBJECTIVES

At the end of the lesson, learners are able to:

- (Knowledge)** recall the rules for aligning decimal points in addition/subtraction and placing the decimal point in products/quotients;
- (Skills)** add, subtract, multiply, and divide decimals, and solve multi-step routine and non-routine problems;
- (Values)** show accuracy and honesty when handling money in real transactions.

ACTIVITY — “*Tindahan Day.*” Groups receive a price list (rice ₱52.75/kg, egg ₱8.50, cooking oil ₱98.25). Each group “buys” items totaling a given budget of ₱200 and computes their change using play money.

ANALYSIS — Guide questions: How did you line up the numbers when adding prices? What happens to the decimal point when you multiply $₱52.75 \times 3$? When you divide a bill equally among 4 friends, where does the decimal point go?

ABSTRACTION — Formalize the rules:

- **Add/Subtract:** align decimal points, then add/subtract as with whole numbers.
- **Multiply:** multiply as whole numbers; the product has as many decimal places as the *total* in the factors.
- **Divide:** move the divisor’s decimal to make it a whole number, move the dividend’s the same number of places, then divide.

Worked examples: $12.6 + 4.85 = 17.45$; $9.3 - 5.74 = 3.56$; $5.2 \times 0.6 = 3.12$; $4.5 \div 0.5 = 9$.

APPLICATION — Problem: “*Aling Rosa bought 2.5 kg of fish at ₱180.00/kg and paid with a ₱500 bill. How much change did she receive?*” (Answer: $2.5 \times 180 = ₱450$; $500 - 450 = ₱50.00$.)

FORMATIVE ASSESSMENT / ASSIGNMENT

3-item exit pass (one per operation). Assignment: bring one real receipt and verify two computations on it.

Lesson 2 · Adding and Subtracting Fractions and Mixed Numbers

NA

6NA-IC-3

Subject Matter	Addition and subtraction of fractions and mixed numbers (like and unlike denominators)
Time	2 sessions (100 min)
Materials	fraction strips/tiles, paper circles, recipe cards

LEARNING OBJECTIVES

(K) explain why fractions need a common denominator before adding/subtracting; **(S)** add and subtract fractions and mixed numbers, regrouping when needed, and express answers in lowest terms; **(V)** value teamwork and patience in step-by-step work.

ACTIVITY — “Sharing the Bibingka.” Using paper circles cut into halves, thirds, and sixths, groups model $\frac{1}{2} + \frac{1}{3}$ and physically find an equivalent shared piece.

ANALYSIS — Why can’t we just add the numerators and denominators? What single size of piece can represent both halves and thirds? (sixths)

ABSTRACTION — Steps: find the LCD → rename to equivalent fractions → add/subtract numerators → simplify. Example: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. Mixed numbers: $3\frac{1}{4} - 1\frac{3}{4} = 1\frac{1}{2}$ (regroup 1 whole as $\frac{4}{4}$).

APPLICATION — “A recipe needs $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar. How many cups of dry ingredients in all?” (Answer: $4\frac{1}{4}$ cups.)

FORMATIVE ASSESSMENT / ASSIGNMENT

Think-pair-share board work (2 items). Assignment: write one real situation at home that uses fraction addition.

Lesson 3 · Multiplying and Dividing Fractions NA 6NA-Id-4

Subject Matter	Multiplication and division of fractions, whole numbers, and mixed numbers
Time	2 sessions (100 min)
Materials	grid paper, fraction tiles, string/ribbon for measuring

LEARNING OBJECTIVES

(**K**) state that “of” means multiply and that dividing by a fraction means multiplying by its reciprocal; (**S**) multiply and divide fractions/mixed numbers and solve related problems; (**V**) appreciate precision in measuring and sharing.

ACTIVITY — “Half of a Half.” On grid paper, learners shade $\frac{1}{2}$ of a region, then shade $\frac{1}{3}$ of the shaded part and name the result ($\frac{1}{6}$).

ANALYSIS — What operation does “ $\frac{1}{3}$ of $\frac{1}{2}$ ” suggest? When we cut 3 metres of ribbon into $\frac{1}{4}$ metre pieces, are we multiplying or dividing?

ABSTRACTION — **Multiply:** multiply numerators, multiply denominators, simplify. Convert mixed numbers to improper first. **Divide:** multiply by the reciprocal of the divisor. Examples: $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$; $3 \div \frac{1}{4} = 3 \times 4 = 12$.

APPLICATION — “A $3\frac{1}{2}$ -metre ribbon is cut into $\frac{1}{2}$ -metre pieces. How many pieces?” (Answer: $\frac{7}{2} \div \frac{1}{2} = 7$ pieces.)

FORMATIVE ASSESSMENT / ASSIGNMENT

Quick board relay (4 items). Assignment: draw a model showing $\frac{2}{5}$ of 10.

Lesson 4 • GCF and LCM

NA**6NA-1e-5**

Subject Matter	Greatest Common Factor and Least Common Multiple of 2–3 numbers; applications
Time	2 sessions (100 min)
Materials	counters, factor trees on cards, Venn diagram templates

LEARNING OBJECTIVES

(K) distinguish factors from multiples and GCF from LCM; **(S)** find the GCF and LCM using listing, prime factorization, and continuous division, and apply to problems; **(M)** demonstrate orderliness and logical thinking.

ACTIVITY — “Goody Bags.” Given 24 candies and 36 biscuits, groups find the greatest number of identical goody bags they can make with nothing left over.

ANALYSIS — What number of bags works for BOTH items? Is that the largest such number? How is this different from finding when two blinking lights light up together (LCM)?

ABSTRACTION — Demonstrate three methods (listing, prime factorization, continuous division).

Example: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$. **GCF** = $2^2 \times 3 = 12$; **LCM** = $2^3 \times 3^2 = 72$. GCF → “largest equal groups”; LCM → “next time events coincide / common denominator.”

APPLICATION — “Bus A leaves every 12 min, Bus B every 18 min. If both leave at 7:00 a.m., when do they next leave together?” (Answer: $\text{LCM}(12, 18) = 36 \rightarrow$ **7:36 a.m.**)

FORMATIVE ASSESSMENT / ASSIGNMENT

Venn-diagram seatwork (GCF & LCM of 16 and 20). Assignment: find GCF/LCM of your & a sibling’s age.

Lesson 5 • Exponents and the GEMDAS Rule

NA

6NA-If-6

Subject Matter	Reading/writing numbers in exponential form; order of operations (GEMDAS)
Time	2 sessions (100 min)
Materials	square/cube blocks, operation cards, mini-whiteboards

LEARNING OBJECTIVES

(K) identify base and exponent and state the GEMDAS order; (S) write/evaluate numbers in exponential form and simplify expressions using GEMDAS; (V) value following correct procedures and rules.

ACTIVITY — “Folding Paper.” Learners fold paper repeatedly and record the number of regions (2, 4, 8, 16...) leading to 2^1 , 2^2 , 2^3 , 2^4 .

ANALYSIS — What does the small raised number tell us? In $2 + 3 \times 4$, do we add or multiply first — and why does the order change the answer?

ABSTRACTION — **Exponential form:** b^n means b used as a factor n times (e.g. $5^3 = 5 \times 5 \times 5 = 125$).

GEMDAS: Grouping → Exponents → Multiply/Divide (left-right) → Add/Subtract (left-right).

Example: $2 + 3 \times 4^2 = 2 + 3 \times 16 = 2 + 48 = 50$.

APPLICATION — Evaluate $(6 + 4) \div 2 + 3^2 = 10 \div 2 + 9 = 5 + 9 = 14$.

FORMATIVE ASSESSMENT / ASSIGNMENT

Mini-whiteboard flash (3 expressions). Assignment: create one GEMDAS expression whose answer is 20.

Lesson 6 • Tessellation MG 6MG-Ig-7

Subject Matter	Identifying shapes that tessellate; creating tessellation patterns (Filipino designs)
Time	1–2 sessions (50–100 min)
Materials	pattern blocks/cutouts (triangles, squares, hexagons, pentagons), colored paper, samples of <i>banig</i> weaving

LEARNING OBJECTIVES

(K) define tessellation and explain why some shapes tessellate and others do not (angles around a point sum to 360°); **(S)** identify tessellating shapes and create an original tessellation; **(V)** appreciate geometry in Filipino art and weaving (*banig*, *habi*).

ACTIVITY — “*Tiling the Floor.*” Groups try to cover a sheet with no gaps/overlaps using squares, triangles, hexagons, and regular pentagons; they record which ones work.

ANALYSIS — Which shapes left gaps? Look at the angles meeting at a point — what must they add up to? (360°) Why does a regular pentagon (108°) fail?

ABSTRACTION — A **tessellation** is a repeating pattern of shapes covering a plane with no gaps or overlaps. Regular tessellations exist for the equilateral triangle (60°), square (90°), and regular hexagon (120°) because their angles divide 360° evenly. Connect to Filipino *banig* mats and *habi* weaving.

APPLICATION — Each learner designs an original tessellation inspired by a Filipino weaving pattern and labels the repeating shape.

FORMATIVE ASSESSMENT / ASSIGNMENT

Gallery walk + rubric (correctness, no gaps, creativity). Assignment: photograph/sketch one tessellation found at home or in your barangay.